

# Cross-City Comparison of Crime-Prediction Models: A Scale Confound in Aggregate Poisson Likelihood, with a Four-City Demonstration

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## Abstract

**Objectives.** Aggregate event-weighted Poisson predictive log-likelihood (PLL) is widely reported when comparing count-prediction models across populations of differing density (e.g., crime prediction across cities). This paper asks whether the metric is fit for that purpose and proposes an alternative metric pair for cross-population evaluation.

**Methods.** The paper documents, algebraically and empirically, that aggregate PLL is confounded by zero-rate composition: the per-event log-likelihood ceiling,  $\max_y \log p(y | \mu) \approx -\frac{1}{2} \log(2\pi\mu)$ , becomes more negative as the predicted mean grows, regardless of model quality. The confound is demonstrated on a four-city LightGBM crime-prediction pipeline (Boston, Leeds, Birmingham, London) using publicly available incident data at H3-resolution-9 monthly granularity, with 5-seed ensembling and per-(city, type) honest decomposition.

**Results.** Three different cities take rank-1 under three metrics — Birmingham on aggregate PLL, Boston on `nonzero_pll_mean`, London on `calibration_ratio` — and no city holds rank across all three. Denser populations appear systematically worse on aggregate PLL even when their honest per-event predictive likelihood and calibration are comparable or better. Four configuration-specific gradient-space negative results are also reported; the divergence-detection diagnostic against which they were tested measures sparsity rather than predictive pathology.

**Conclusions.** The paper proposes `nonzero_pll_mean` and `calibration_ratio` as the primary cross-population evaluation pair, with `zero_rate` as descriptive context. Implications for cross-city benchmarking of crime-prediction models are discussed.

**Keywords:** cross-population evaluation; Poisson regression; calibration; gradient boosting; sub-municipal crime prediction; negative results; H3 spatial indexing.

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## 1. Introduction

### 1.1 Motivation

Crime concentrates at small spatial scales. Weisburd (2015) names this regularity the “law of crime concentration”: across eight U.S. cities, roughly 50 percent of crime occurs at 4–5 percent of street segments, with the concentration stable over

time. Earlier and later work documents the same pattern in additional cities and crime types (Sherman, Gartin, & Buerger, 1989, Minneapolis; Groff, Weisburd, & Yang, 2010, Seattle longitudinal). Spatial concentration motivates fine-grained predictive modeling at sub-municipal scales (Mohler et al., 2011; Mohler, 2014; Wheeler & Steenbeek, 2021). One operational application — hot-spot policing — has been evaluated in a systematic review and meta-analysis (Braga et al., 2019), which finds crime-reduction effects at targeted micro-places. Kounadi et al. (2020) provide a systematic review of the broader spatial crime-forecasting literature.

A growing body of cross-city machine-learning validation work compares predictive performance across multiple jurisdictions (Hardyns & Khalfa, 2023; Khalfa, Snaphaan, Ristea, Kounadi, & Hardyns, 2025; Kronkvist et al., 2025). Adjacent criminological work examines the variability of crime concentration across spatial units within cities (Steenbeek & Weisburd, 2016) and provides nonparametric methods for testing similarity between spatial point patterns across areas (Andresen, 2009). The specific metric varies with the modeling framing: predictive accuracy index (PAI; Chainey, Tompson, & Uhlig, 2008) and hit rate are commonly reported in operational predictive-policing comparisons, while event-weighted predictive log-likelihood — an aggregation of the per-observation logarithmic score, which is a strictly proper scoring rule (Gneiting & Raftery, 2007) — is commonly reported in count-modeling and forecasting-evaluation comparisons. A methodological literature on assessing predictive count models in absolute terms is well-developed (Czado, Gneiting, & Held, 2009; Cameron & Trivedi, 2013). The paper addresses a methodological prerequisite that, to the authors’ knowledge, has not been explicitly treated in this comparison literature: whether the aggregate Poisson predictive log-likelihood is comparable across populations of differing zero-rate composition.

## 1.2 Scope and contributions

This paper is a methodological note. This paper makes two contributions:

1. **Scale-confounded aggregate PLL.** The paper shows, both algebraically and through a four-city empirical demonstration, that event-weighted Poisson predictive log-likelihood is confounded by zero-rate composition. The confound is large enough to reshuffle the apparent ranking of cities depending on which evaluation metric is used: in the four-city corpus, three different metrics produce three different rank-1 cities, and no city holds rank across all three. This paper proposes `nonzero_pll_mean` and `calibration_ratio` as the primary cross-population pair, with `zero_rate` reported as descriptive context.
2. **A catalog of four configuration-specific gradient-space negative results.** In the same LightGBM-based pipeline (applied uniformly to all 36 (city, type) cells across the four cities), the analysis tested four interventions targeted at a divergence-detection diagnostic that fired specifically on heavy-

tailed spatial concentration of shoplifting events under monthly aggregation: Tweedie loss at variance power 1.5, negative binomial loss at three fixed dispersion settings tested under default and two reduced learning rates, training-target winsorization at three percentiles, and single-type learning-rate reduction by factor of 5. Three of the four (Tweedie, NB, single-type lr-reduction) failed to improve predictions in this configuration; the fourth (winsorization) gave a mixed result — viable at the 99.9 percentile but distortional at lower percentiles. On retrospective inspection, the divergence diagnostic itself was measuring spatial sparsity rather than predictive pathology — a finding the paper reports alongside the four interventions in §4.4 — so none of the four interventions was adopted into the production pipeline. This paper reports the negative results explicitly as configuration-specific, not as categorical rejections of the loss families or techniques involved. They are offered as observations for practitioners working in similar regimes.

The four-city LightGBM application that supplies the empirical demonstrations for both contributions is described in §2–§3. The paper does not claim cross-jurisdictional generalization for the LightGBM architecture; the four-city study is small ( $n = 4$ ), and the four cities are all high-income OECD English-speaking common-law jurisdictions with similar policing institutions and data-recording traditions. The four-city corpus is appropriate for *illustrating* the metric confound but not for *establishing* architectural generalization. This paper makes this scope explicit throughout.

A subsidiary observation on whether the existing UK Home Office 14-category taxonomy reduces stably under empirical clustering is reported in Appendix A. The authors are careful to characterize this as a finding about the reducibility of an existing taxonomy, not about taxonomy generalization across jurisdictions.

### 1.3 Prior work this builds on

This study is part of a multi-jurisdiction safety-analytics research program. Earlier work in the program established a LightGBM prediction pipeline on Boston Police Department weekly data with engineered features. The present paper builds on that pipeline using a different feature set and evaluation protocol; the earlier results are not used as a comparison baseline. Earlier phases of the program also established that several statistical overlays — Bühlmann-Jewell credibility (Bühlmann & Gisler, 2005), negative binomial, and isotonic calibration (Niculescu-Mizil & Caruana, 2005) — did not improve on the base LightGBM model in the Boston-only setting, and that defensive layers (anomaly flagging, event probability calibration) operate orthogonally to core model fit. The present paper is methodologically focused on the cross-population evaluation problem rather than on extending the predictive validation across jurisdictions.

## 2. Data

### 2.1 Event data

**Boston.** Boston Police Department incident data, January 2023 through April 2026 (40 months, including one partial final month; see §3.4), obtained through the Analyze Boston open-data portal (publicly available; no data-use agreement). Each event is timestamped at daily resolution with latitude and longitude, classified under 58 NIBRS sub-categories (Federal Bureau of Investigation, 2021). Total: 115,532 events.

**UK cities.** data.police.uk public crime data (UK Home Office n.d.), March 2023 through February 2026 (36 months) for Leeds, Birmingham, and London. Each event is timestamped at monthly resolution with latitude and longitude, classified under one of 14 Home Office categories. Totals: Leeds 244,991; Birmingham 483,354; London 1,691,514. The three cities were selected as a convenience sample of large English cities spanning roughly a seven-fold event-volume range; this paper does not claim the selection is representative of UK cities broadly (consistent with the §1.2 scope limitation). The UK study window covers the 36 most recent months available at extraction time (early 2026); the pipeline adopted a 36-month window for cross-city comparability and selected the most recent contiguous span ending February 2026, which begins in March 2023. The two-month start-date offset relative to Boston (January, 2023) is therefore a property of the window-selection choice rather than a Boston–UK temporal alignment. Training-bin counts are aligned across cities at 24 bins per city (§3.3), at the cost of slightly different absolute calendar windows and slightly different validation/test bin counts (Boston: 7 validation, 8 test; UK cities: 6 validation, 6 test). Under the interleaved year-3 split described in §3.3 the test bins fall in alternating year-3 calendar months: Boston test bins land in January, March, May, July, September, November 2025 and January, March 2026 (eight test bins); UK test bins land in March, May, July, September, November 2025 and January 2026 (six test bins). The six bin-months in which all four cities simultaneously contribute test data are therefore March, May, July, September, November 2025 and January 2026 (the odd-month subset within a March-2025-to-January-2026 calendar span); Boston contributes two additional non-overlap test bins (January 2025 and March 2026).

**data.police.uk anonymization caveat.** Per data.police.uk publication policy, UK event coordinates are not the actual incident location but are snapped to one of a published set of “anonymous map points” within approximately 100–200 meters of the incident. At the H3 resolution-9 scale used here (cell radius approximately 175 meters; §2.3), this snapping introduces non-trivial measurement error in the cell-of-occurrence assignment for UK events. Boston coordinates, by contrast, are not subject to a comparable snap-point procedure. Cell-level attribution noise is therefore likely larger for the three UK cities than for Boston. This paper does not separately quantify or correct for this attribution noise in the present analysis; it is one limitation of the cross-city setup and is

consistent with the broader institutional-homogeneity caveat in §5.2.

**Granularity constraint.** All UK events are published with monthly-only timestamps: 100 percent of events in each UK city have day-of-month equal to 1. This is an architectural property of the data.police.uk publication protocol, not a data-quality issue. It imposes a hard ceiling of 36 temporal bins for the 3-year UK study period. To make Boston comparable, the analysis re-aggregate Boston’s daily events to monthly granularity (40 bins for Boston’s 40-month window).

**Test-window context.** Test bins for all four cities fall in 2025–2026. A widely-reported concurrent trend in UK retail crime over this window is a cost-of-living-driven rise in shoplifting volumes; this paper does not independently verify or model this effect, and the paper reports results without conditioning on cost-of-living covariates. The empirical results below are conditional on this specific test window; the paper reports a sensitivity analysis on an alternative test window (Window A) in Appendix C.

## 2.2 Covariates

This paper uses Kontur (2024) population estimates at H3 resolution 8, joining to event cells via r8-parent lookup and attributing parent population divided by seven to each r9 child cell (the H3 hierarchical containment relation: each non-pentagonal r8 cell contains exactly seven r9 child cells; none of the four study areas overlap the 12 globally-located H3 pentagonal cells, so the divide-by-7 attribution is valid for all cells in this study). The  $/7$  attribution implicitly treats population as uniformly distributed within each r8 parent ( $\sim 0.74 \text{ km}^2$ ); attribution error from within-parent population gradients is one source of feature noise this paper does not separately quantify. Coverage — defined as the proportion of occupied cells (cells with at least one event over the study window) for which a non-zero Kontur population estimate is available — is 99.9 percent for Boston and 100 percent for each UK city. Cells with zero attributed population (rare; predominantly non-residential cells where events occur but no Kontur residential estimate is available, including a small fraction of Boston cells) are handled by computing the feature as  $\log(\text{pop} + 1)$  rather than  $\log(\text{pop})$  to keep the value defined; the  $+1$  smoothing is negligible relative to typical per-cell populations of order  $10^2$ – $10^3$ . This feature represents log-population in the cell on the same scale as a GLM-style Poisson offset, but in the LightGBM implementation it enters as a regular feature in the 14-feature stack — the booster learns a non-linear function of `log_pop_kontur` through its tree splits rather than constraining it to enter as a fixed log-linear offset (which a strict Poisson offset would require). The pipeline retains the variable name `log_pop_kontur` and refer to it informally as a population-density baseline below.

### 2.3 Spatial indexing

All events are indexed to H3 resolution-9 cells (approximately 0.105 km<sup>2</sup> average area, or approximately 10.5 hectares; characteristic hexagon edge length ~174 m, vertex-to-opposite-vertex diameter ~349 m, edge-to-opposite-edge width ~302 m; Uber Technologies, n.d.). Boston’s study area yields 1,031 occupied cells (cells containing at least one event over the study window); Leeds 1,100; Birmingham 2,446; London 2,997. The H3 resolution-9 scale was chosen for practical indexability at scale across all four cities. The crime-concentration literature documents concentration primarily at the finer street-segment scale (Weisburd, 2015; Groff, Weisburd, and Yang, 2010); an r9 cell typically contains multiple street segments, and concentration at r9 is expected to be present but may be attenuated relative to documented street-segment findings. This paper does not claim that H3 r9 cells are equivalent units to street segments; the choice trades finer spatial granularity for cross-city indexability and computational tractability, and the effect of this trade-off on cross-city predictive comparisons is not separately quantified (see Rummens & Hardyns (2021) for direct evidence that spatiotemporal resolution choices shift predictive-policing model performance, in some cases substantially). Recent comparative work by Khalfa & Hardyns (2025) finds no substantial prediction-performance differences between rectangular and hexagonal grid shapes at matched resolutions in crime ML prediction.

The H3 cell area (~0.105 km<sup>2</sup>, ~10.5 hectares) typically contains multiple establishments — at urban densities, an r9 cell in a commercial district can contain dozens of retail premises, restaurants, and other facilities; in residential areas it more typically contains tens to a few dozen addresses; this paper does not separately quantify per-cell facility counts in this study. Cell-level concentration is therefore not direct evidence of facility-level concentration, an important distinction the paper returns to in §4.2.

**Study-area populations.** Population within the H3-r9 study extent (the union of occupied cells), as attributed by Kontur: Boston ~580,000, Leeds ~750,000, Birmingham ~1,050,000, London ~6,400,000. These are the relevant population denominators for prediction; the cities’ total municipal populations (Boston ~675K; Leeds ~810K, Leeds City Council district; Birmingham ~1.14M, Birmingham UK city; London ~9M, Greater London administrative area) include territory outside the H3-r9 extent. The implied study-area-to-municipal coverage fractions are approximately Boston 86 percent, Leeds 93 percent, Birmingham 92 percent, London 71 percent; the lower London figure reflects the larger Greater London footprint relative to the occupied-cell extent.

### 2.4 Crime taxonomies

**Boston: BPD K5 grouping.** Boston’s NIBRS sub-categories (Federal Bureau of Investigation 2021) are grouped into 5 bands: band\_0 “Nuisance/Disorder”, band\_1 “Theft/Fraud”, band\_2 “Serious Property”, band\_3 “Drugs/Weapons”, band\_4 “Violent”. The grouping was defined ex ante by BPD analysts collabo-

rating with the project, prior to the analyses reported here; the analysis treats it as the operational taxonomy for this study without claiming it is a published official BPD scheme. The full sub-category-to-band mapping is in the project repository.

**UK: native Home Office categories.** data.police.uk publishes events in 14 categories: anti-social-behaviour, bicycle-theft, burglary, criminal-damage-arson, drugs, other-crime, other-theft, possession-of-weapons, public-order, robbery, shoplifting, theft-from-the-person, vehicle-crime, and violent-crime.

**Boston K5 to UK mapping.** No canonical mapping exists between Boston’s K5 bands and the UK 14-category schema. The mapping below is used in Appendix A.5 to compute cross-taxonomy ARI between the chosen mapping and empirical UK clustering; it is the author’s best-effort heuristic.

- band\_4 “Violent” → UK violent-crime + robbery. (UK robbery is published as a separate top-level category in data.police.uk but is grouped with violence-against-the-person in UK Home Office statistical reporting because it involves the use or threat of force; the analysis follows that convention here.)
- band\_3 “Drugs/Weapons” → UK drugs + possession-of-weapons.
- band\_2 “Serious Property” → UK burglary + vehicle-crime + criminal-damage-arson.
- band\_1 “Theft/Fraud” → UK other-theft + shoplifting + theft-from-the-person + bicycle-theft. (The K5 band name includes “fraud,” but the data.police.uk 14-category schema does not include a fraud category — UK fraud is reported separately by Action Fraud — so the K5 band’s fraud sub-component has no symmetric UK counterpart in this corpus.)
- band\_0 “Nuisance/Disorder” → UK anti-social-behaviour + public-order + other-crime. (**other-crime** is the data.police.uk residual category for offences not classified under the other 13 headings; the analysis groups it with the BPD K5 nuisance/disorder band on the grounds that both serve as residual buckets, while acknowledging the substantive composition of the two residuals may differ.)

This mapping accounts for all 14 UK Home Office categories. Boston’s band\_3 falls below the 10,000-event threshold described in §3.4 and is therefore excluded from the §4 results in any case; the mapping is documented for completeness and reproducibility. The mapping should not be confused with a substantive equivalence claim — it is the author’s best-effort heuristic, and Boston-UK per-type comparisons are reported only as approximate and qualitatively interpretable.

**Why per-city native taxonomies.** No accepted single taxonomy exists across US and UK jurisdictions. The UK 14 Home Office categories are the published data.police.uk reporting categories. The BPD K5 bands are an internal aggregation defined for this project as described above. The pipeline retains each city’s working taxonomy throughout, accepting the resulting loss of like-for-like cross-Boston–UK per-type comparison.



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### 3. Methods

#### 3.1 Model

Per city and per native crime type, the pipeline trains an independent LightGBM Booster with Poisson objective. This paper uses LightGBM (Ke et al., 2017) without separate benchmarking against comparable gradient-boosted tree implementations (Chen & Guestrin, 2016; Friedman, 2001). GBDT-family models have been used in recent machine-learning-based crime prediction research (Khalifa & Hardyns, 2025; Khalifa, Theinert, & Hardyns, 2025); Wheeler & Steenbeek (2021) use random forests, another tree-ensemble family. Hyperparameters were inherited from an earlier Boston-specific configuration developed during the project’s Wave 1 phase and applied unchanged to the four-city analysis: learning rate 0.05, num\_leaves = 31, L1 and L2 regularization both at 0.1, feature\_fraction = 0.9, bagging\_fraction = 0.8, num\_iterations = 500, early\_stopping\_rounds = 30 (i.e., training halts if the validation-set Poisson objective fails to improve for 30 consecutive boosting rounds). The paper flags the inherited-from-Boston configuration as a methodological caveat: hyperparameters were not retuned per city, so per-city PLL differences include a (small but non-zero) component attributable to the use of the Boston-developed configuration on UK data; this paper does not quantify this component in the present paper.

#### 3.2 Features

The final 14-feature stack used in this paper:

| Feature                 | Definition                                                                                                                                                                       |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| log_pop_kontur          | Log-population in cell (Kontur r8 parent divided by 7); population-density baseline (see §2.2 for implementation note: enters as a regular feature, not a strict Poisson offset) |
| lag_1, lag_3, lag_6     | Count in same (cell, type) at lag 1, 3, 6 months                                                                                                                                 |
| rm_2, rm_4, rm_6, rm_12 | Rolling means of count in same (cell, type) at horizons 2, 4, 6, 12 months                                                                                                       |
| quarter                 | Month-of-year quarter (categorical 1–4)                                                                                                                                          |
| trend_slope_6           | Linear slope of previous 6 months’ counts per cell-type                                                                                                                          |

| Feature                           | Definition                                                                                                                                    |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <code>nb_sum_1t</code>            | Sum of <code>lag_1</code> over k-ring-1 neighbor cells of the same type (the six immediately adjacent r9 hexagons surrounding the focal cell) |
| <code>other_types_sum_1t</code>   | Sum of other types' <code>lag_1</code> at the same cell                                                                                       |
| <code>other_types_sum_rm_4</code> | Sum of other types' 4-month rolling mean at the same cell                                                                                     |
| <code>type_rank_in_cell</code>    | Static historical rank of this type at this cell                                                                                              |

The 14-feature stack was arrived at by ablating from a 22-feature expansion. The eight dropped features were two long-horizon rolling means (`rm_18`, `rm_24`; both largely zero on bins within the first 24 months of the training window because they require a full 18- or 24-month look-back), two long-horizon lags (`lag_9`, `lag_12`), one neighborhood feature (`nb_count_1t`, the count of same-type events in the focal cell's k-ring-1 neighborhood, which proved redundant with the retained `nb_sum_1t`), one neighborhood-diversity feature (`nb_diversity_1t`), one cell-level concentration feature (`spatial_concentration`), and one sin/cos seasonal pair (`month_sin`, `month_cos`, treated as a single cyclic-encoding feature, made redundant by the retained `quarter` feature). Selection was driven by LightGBM split-importance gain in a London-trained model, with features contributing less than 2 percent of total gain dropped (the appendix discusses in §C.4 the consequence that this step is therefore not strictly external-to-London). All lag and rolling features apply `shift(1)` before aggregation so current-bin data never feeds its own prediction. The analysis verifies by direct inspection that at the first available bin per city, all `lag_*` and `rm_*` features equal zero — a logical consequence of `shift(1)` with no prior history. The 14-feature stack is substantially simpler than the earlier Boston-specific stack used in prior project phases, which included holiday indicators, weather proxies, and sub-monthly features not materializable from data.police.uk.

### 3.3 Train/validation/test split

This paper uses an **interleaved year-3 split**:

- Train: the first 24 monthly bins (approximately years 1–2).
- Validation: year-3 bins with  $(\text{bin} - 24)$  even (used for early stopping).
- Test: year-3 bins with  $(\text{bin} - 24)$  odd (held out).

For UK cities (36 bins, year-3 = bins 25–36, twelve year-3 bins): under the  $(\text{bin} - 24)$  parity rule, bins 26, 28, 30, 32, 34, 36 are validation (six) and bins 25, 27, 29, 31, 33, 35 are test (six). For Boston (39 bins after partial-bin exclusion described in §3.4, year-3 = bins 25–39, fifteen year-3 bins): under the same rule,

bins 26, 28, 30, 32, 34, 36, 38 are validation (seven) and bins 25, 27, 29, 31, 33, 35, 37, 39 are test (eight). Boston therefore has 24 train / 7 validation / 8 test bins; each UK city has 24 / 6 / 6. By construction, validation and test bins alternate across the entire year-3 window: each block of trend or seasonality longer than two consecutive bins is straddled by both groups, mitigating the contiguity-induced asymmetry that arises under fractional contiguous splits (where, for example, a 75/25 split would put the entire validation block at one end of year-3 and leave the test block at the other). Note that the validation and test sets occupy **disjoint calendar months** under this design — validation gets year-3’s even-indexed calendar offsets and test gets the odd-indexed ones — so the design is not a “matched-month” split. Rather, it ensures both groups span the full year-3 window so that any single concentrated drift, holiday, or seasonal event affects both groups in proportion. This paper presents this split as a design choice, not as a contribution; alternative chronological split designs would produce different results, and this paper does not claim interleaved splits are uniformly preferred to fractional splits in count-model evaluation.

### 3.4 Category filtering and partial-bin exclusion

**Category filtering.** Categories with fewer than 10,000 total events over the study window are excluded and documented. The 10,000-event threshold was chosen ex ante for cross-city consistency and to ensure approximately 250–280 events per monthly bin per category averaged over the 40-month Boston and 36-month UK windows, sufficient for stable per-month conditional means. At the present extraction, the threshold binds only on Boston **band\_3** “Drugs/Weapons” (8,237 events), which is therefore reported but not modeled. The analysis did not perform a sensitivity analysis on the threshold value.

**Partial-bin exclusion.** Raw event data from BPD terminates at an arbitrary calendar date determined by extraction time. In the working copy, Boston’s last monthly bin (April, 2026) contained only the first 10 days of data. Including this bin in the test set would have introduced a downward bias in **calibration\_ratio** proportional to the truncation — observed events covered roughly the first third of the month while the model predicted a full month — attributable entirely to incomplete coverage rather than to any change in the underlying event rate. This was an extraction-freshness artifact specific to working with a live BPD endpoint at a fixed snapshot date, not a methodological insight. The pipeline detects and excludes bins with fewer than 28 days of coverage. UK cities report all events at day-of-month = 1 by aggregation at source, so the check is unneeded for them. This paper does not present partial-bin handling as a generalizable methodological recommendation; the paper presents it as documentation of a specific extraction-pipeline artifact and its remediation in the setup, which other groups working with similar live event-data feeds may want to check for.

### 3.5 Metrics

The analysis computes Poisson predictive log-likelihood (PLL) on the held-out test set per (city, type). For a single observation with observed count  $y$  and predicted mean  $\mu$ ,

$$\log p(y \mid \mu) = y \log \mu - \mu - \log(y!)$$

This paper reports four summary metrics per city; Cameron & Trivedi (2013) provide the textbook treatment of Poisson and negative-binomial regression and their evaluation that the analysis follows throughout.

1. **Event-weighted mean PLL** (the scale-confounded aggregate, retained for comparison and discussed in §4): mean PLL weighted by  $n\_test\_events$  per type.
2. **nonzero\_pll\_mean**: mean PLL over test observations where  $y > 0$ . Captures model performance conditional on event occurrence. For (city, type) pairs in which no test observation has  $y > 0$ , **nonzero\_pll\_mean** is undefined; this case does not arise for any of the 36 (city, type) cells reported in this paper but is flagged for practitioners working with smaller test sets.
3. **calibration\_ratio**:  $\bar{y}/\bar{\mu}$  over the full test set. 1.0 indicates a perfectly calibrated mean; values above 1 indicate under-prediction, below 1 over-prediction.
4. **zero\_rate**: fraction of test observations where  $y = 0$ . Reported as diagnostic context for the aggregate PLL.

Per-city **nonzero\_pll\_mean** is computed by pooling all (cell, month, type) observations within a city, filtering to  $y > 0$ , and taking the unweighted mean of the per-observation Poisson log-pmf. Bootstrap CIs at the city level use a type-stratified resampler (each (city, type) cell resampled to its original size, then pooled across types) to preserve each city’s modeled-type composition while operating on the pooled per-observation metric. The type-equal-weighted alternative (arithmetic mean across types of the per-(city, type) **nonzero\_pll\_mean**, rather than the pooled mean over per-observation values) is reported separately in Table B.2.

This paper reports 95 percent bootstrap confidence intervals for **nonzero\_pll\_mean** and **calibration\_ratio** per (city, type) cell using 1,000 resamples, the standard recommended floor for stable 95 percent percentile-based bootstrap CIs of mean-like quantities (Davison & Hinkley, 1997, §5.2); these are tabulated in Appendix B.

The bootstrap resamples observations independently within each (city, type) cell rather than clustering on H3-cell identity. Because observations within a single H3 cell are positively autocorrelated across months, this independent resampling underrepresents within-cell dependence and produces confidence intervals that are slightly narrower than a cluster bootstrap on H3 cells would produce. The pipeline retains the independent resample for tractability; because

it under-represents within-cell dependence, the resulting intervals are likely slightly anti-conservative (i.e., they may under-cover the nominal 95 percent level). Readers should interpret them as approximate precision ranges rather than as exact nominal frequentist coverage (Cameron & Miller, 2015).

The derivation motivating use of the four metrics is presented in §4.1.

**Training-time adjustments.** The pipeline applies exponential time-decay weights to training observations:  $w_i = \exp(-\lambda(t_{\max} - t_i))$  with  $\lambda = 0.15$ , where  $t_i$  is the bin index of observation  $i$  and  $t_{\max}$  is the most recent training bin. With training bins indexed  $1, \dots, 24$  and  $t_{\max} = 24$ , this assigns weight 1.0 to the most recent training bin and weight  $\exp(-0.15 \times 23) \approx 0.032$  to the oldest. Validation observations occur after  $t_{\max}$  (so  $t_{\max} - t_i$  is negative, which would give weights greater than 1); the analysis caps their weights at 1.0. The value  $\lambda = 0.15$  was set during earlier Boston-only pipeline development and retained in the present study without re-tuning; it corresponds to a half-life of  $\ln(2)/0.15 \approx 4.6$  bins (months), which judged appropriate for monthly crime data exhibiting moderate temporal stationarity. The analysis did not perform a formal sensitivity analysis on  $\lambda$  in the present study.

### 3.6 Reproducibility

The methodological subset of the pipeline — the LightGBM training, ensemble aggregation, audit framework, and reliability classification scripts referenced in §C.2 — will be made available in a public repository upon acceptance, with the canonical Git SHAs marked. The full production pipeline contains commercial-product-specific data ingestion components and is not part of the public release. The values reported in §4.3 (calibration-improvement decomposition), §4.4 (negative-results catalog), and Appendix A (taxonomy clustering metrics) are computed by the scripts listed in §C.2 and will be regenerable from those scripts in the public methodological-subset repository. Boston data is publicly available at [data.boston.gov](http://data.boston.gov); UK data is publicly available at [data.police.uk](http://data.police.uk). No data-use agreements apply. For the three UK cities (Birmingham, Leeds, London), canonical Phase 2 results in Table 1 are from a 5-seed LightGBM ensemble (seeds 1, 2, 3, 4, 5) per city, with predictions averaged at the (cell, month) level before metric computation. Boston canonical Phase 2 predictions are from a single-seed LightGBM run under an earlier pipeline configuration (v15b; LightGBM seed = 42), retained via symlink rather than retrospectively reconstructed; the Boston row of Table 1 is therefore based on single-seed predictions rather than ensemble-averaged predictions. The single-seed-vs-ensemble asymmetry between Boston and the UK cities is a pipeline-assembly artifact that the paper reports explicitly. The methodological argument of §4.1 (scale-confounding of aggregate PLL by zero-rate composition) does not depend on ensemble vs single-seed prediction generation; it depends only on the per-(cell, month)  $\mu$  values, which are well-defined under either. Bootstrap CIs reported throughout this paper use a numpy random generator initialized via `np.random.default_rng(seed=42)` for both dataset- construction randomness (train/validation/test bin assignment) and the

1,000 bootstrap resamples per (city, type) cell. Per-run configuration files and seed values are listed in Appendix C alongside compute-environment details.

## 4. Results

### 4.1 Scale-confounded aggregate PLL: derivation and demonstration

**Algebraic derivation.** For a Poisson observation with mean  $\mu$ , the log-likelihood  $\log p(y \mid \mu) = y \log \mu - \mu - \log(y!)$ , treated as a function of real-valued  $y$ , attains its maximum at  $y = \mu$ . Stirling’s approximation  $\log(y!) \approx y \log y - y + \frac{1}{2} \log(2\pi y)$  applied at this maximum gives

$$\max_y \log p(y \mid \mu) \approx -\frac{1}{2} \log(2\pi\mu).$$

For integer  $y$ , the maximum is attained at the nearest integer to  $\mu$ ; the discrete-vs-continuous correction is bounded by  $\frac{1}{2}(\log \mu - \log(y + 1))$  and is small relative to the  $-\frac{1}{2}\log(2\pi\mu)$  floor for the  $\mu$  values the paper reports.

The Stirling floor for a well-calibrated prediction grows more negative as  $\mu$  grows:  $-0.92$  at  $\mu = 1$ ,  $-1.72$  at  $\mu = 5$ ,  $-2.07$  at  $\mu = 10$ , and  $-2.41$  at  $\mu = 20$  (the formula above is asymptotic and breaks down at  $\mu \lesssim 0.2$  where the continuous- $y$  maximum is dominated by the  $y = 0$  boundary). Zero-count observations contribute only  $-\mu$  to the log-likelihood, which is near zero for small  $\mu$  — these observations are “free” contributions to the aggregate. Non-zero observations contribute at most the floor value (more negative when the model is poorly calibrated), so well-fit models are PLL-bounded above by the Stirling floor. As the zero rate decreases (more non-zero observations per test set), aggregate PLL is pulled toward the non-zero floor. This is a property of the measurement, not of the model.

**Empirical demonstration in the four-city corpus.** Boston’s test set has 74 percent zero observations; London has 61 percent. Under the pipeline configuration described in §3, the four cities have the following aggregate metrics:

**Table 1.** Per-city aggregate metrics. Each metric reports the point estimate followed by a 95 percent bootstrap confidence interval over resampled (cell, month) test observations, 1,000 resamples.

| City       | Types | Test<br>events<br>(sum of $y$ ) | Event-<br>weighted<br>mean<br>PLL | $\text{nonzero\_pll\_mean}$ | $\text{calibration\_ratio}$ | $\text{zero\_rate}$ |
|------------|-------|---------------------------------|-----------------------------------|-----------------------------|-----------------------------|---------------------|
| Boston     | 4     | ~18,000                         | −0.69<br>(−0.70,<br>−0.68)        | −1.97<br>(−2.00,<br>−1.94)  | 0.963<br>(0.95,<br>0.98)    | 0.74                |
| Leeds      | 8     | ~34,000                         | −0.76<br>(−0.77,<br>−0.75)        | −2.06<br>(−2.09,<br>−2.04)  | 0.989<br>(0.98,<br>1.00)    | 0.73                |
| Birmingham | 11    | ~75,000                         | −0.62<br>(−0.62,<br>−0.61)        | −2.07<br>(−2.09,<br>−2.06)  | 0.989<br>(0.98,<br>1.00)    | 0.78                |
| London     | 13    | ~275,000                        | −1.04<br>(−1.04,<br>−1.03)        | −2.02<br>(−2.04,<br>−2.01)  | 0.993<br>(0.99,<br>1.00)    | 0.61                |

**The orderings disagree.** Each metric produces a different rank-1 city. By event-weighted mean PLL: Birmingham > Boston > Leeds > London (Birmingham apparently best, with an end-to-end spread of 0.42 PLL). By  $\text{nonzero\_pll\_mean}$ : Boston > London > Leeds > Birmingham (Boston the rank-1 city on the per-event metric, with the four cities spanning only 0.10 PLL between best and worst central estimates). By  $\text{calibration\_ratio}$ : London > Leeds > Birmingham > Boston (London best-calibrated, Boston worst; Leeds and Birmingham are statistically tied with bootstrap CIs that overlap heavily). No city holds rank across the three metrics, and Birmingham, Boston, and London each rank first on exactly one metric.

Figure 1 visualizes the rank reshuffle. The 13.5-percentage-point difference in zero rate between Boston and London, combined with London’s higher per-cell non-zero  $\mu$ , accounts for the bulk of the 0.35 PLL aggregate gap between the two cities (−0.69 versus −1.04 on event-weighted mean PLL): the decomposition  $(1 - \text{zero\_rate}) \times \text{nonzero\_pll\_mean}$  gives −0.51 for Boston and −0.79 for London, a difference of 0.28 PLL or roughly 80 percent of the observed gap. The remaining  $\sim 0.07$  PLL is contributed by zero-count observations whose predicted means  $\mu$  are not exactly zero (each such observation contributes  $-\mu$  to the PLL). Once comparison is restricted to non-zero observations, the Boston-to-London gap shrinks to 0.05 PLL on the central estimate; on calibration ratio, the ordering reverses entirely (Boston is rank 4 of 4, London is rank 1).

**Figure 1.** Cross-city ranking by metric, illustrating the scale-confound. Panel (a): by event-weighted mean PLL, Birmingham appears best and London worst. Panel (b): by per-event  $\text{nonzero\_pll\_mean}$ , the four cities have central estimates spanning 0.10 PLL, with bootstrap CIs that overlap for the Leeds–Birmingham and Leeds–London pairs but separate cleanly for the Boston pairs. Panel (c):

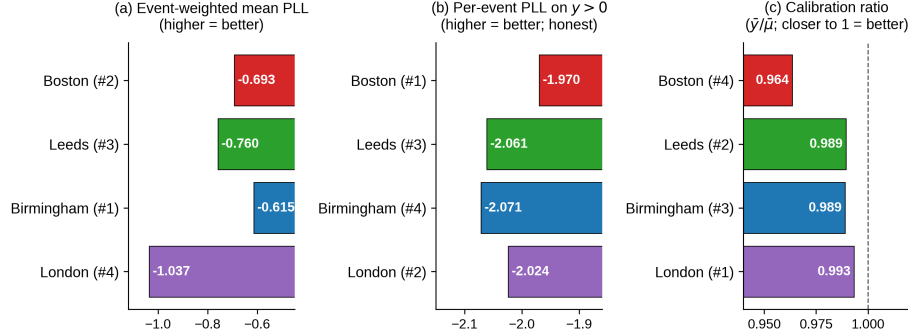


Figure 1: Figure 1

by calibration ratio, Boston is the worst-calibrated of the four cities (#4 of 4) and London the best (#1 of 4). Each metric produces a different rank-1 city; the rank disagreement is a composition effect of differing zero rates (Boston 74 percent, London 61 percent) and per-cell means, not differences in underlying predictive performance.

**Type-level replication.** The same pattern appears in the per-(city, type) decomposition reported in Appendix B and visualized in Figure 2. Across the 36 (city, type) pairs, higher-zero-rate types systematically exhibit more-negative `nonzero_pll_mean`. The three highest-zero-rate pairs are all shoplifting (Birmingham 0.910, Leeds 0.905, London 0.787), which are also among the most-negative on `nonzero_pll_mean`. The lowest-zero-rate pairs (London violent-crime, London anti-social-behaviour) cluster in the  $-2.1$  to  $-2.2$  range despite reflecting large, event-rich populations. The ordering is not strictly monotonic — drugs categories break in the other direction — but the mechanism is clear: types whose events concentrate in few cells produce both high zero rates and high conditional means at non-zero cells, the two conditions that push per-event PLL toward its Stirling floor.

**Figure 2.** Per-(city, type) relationship between zero rate and per-event predictive log-likelihood ( $n = 36$  pairs across four cities). Bubble color encodes city; bubble area is proportional to test event count. The Stirling-floor mechanism is visible at the type level.

**Implication.** Event-weighted mean Poisson log-likelihood is not comparable across populations with different count distributions. Per-event metrics conditioned on observation type (zero vs. non-zero) and direct calibration metrics (observed-over-predicted mean ratio) are recommended, with explicit reporting of zero rate as context. This paper recommends against single-metric cross-population comparisons. For models beyond Poisson — negative binomial, zero-inflated Poisson — analogous decompositions apply with likelihood-specific corrections.



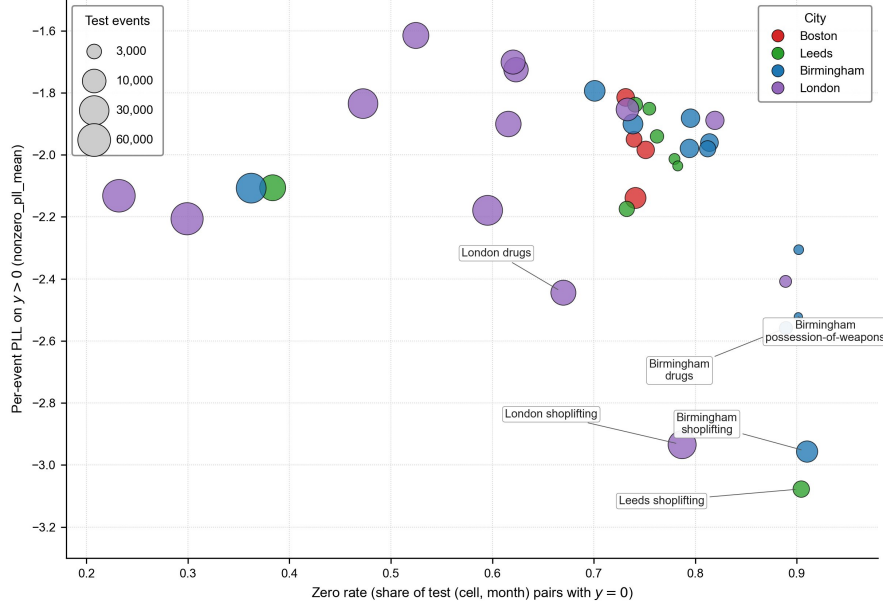


Figure 2: Figure 2

A related metric-definition sensitivity — two conceptually distinct “calibration” formulas (per-row  $\text{mean}(y_i/\mu_i \mid y_i > 0)$ , undefined when no test observation has  $y_i > 0$ ; and aggregate  $\text{mean}(y)/\text{mean}(\mu)$ , the convention adopted throughout this paper for `calibration_ratio`) can produce opposite cross-city orderings on the same data. This paper adopts the aggregate form because (i) it is defined for all (city, type) cells regardless of zero-rate, and (ii) its reading is direct under the §3.5 convention (values  $< 1$  indicate over-prediction). A worked numerical example showing the two formulas disagreeing on the four-city corpus is in the project repository.

#### 4.2 Per-(city, type) patterns under the recommended metrics

The four-city corpus is too small to support inferential cross-city pattern claims. This paper reports three observations as descriptive findings about this specific four-city window, each interpretable through established environmental criminology theory but each requiring replication in future work.

**Observation 1 (smallest cross-UK PLL spread: violent-crime).** Among types present in at least two UK cities, violent-crime has the smallest cross-city spread in `nonzero_pll_mean` ( $\sim 0.10$  PLL between Leeds, Birmingham, and London). Calibration ratios for this type cluster within 0.04. This paper does not describe this as “invariance” — a 0.10 PLL spread corresponds to a substantive likelihood-ratio difference, and bootstrap CIs (Appendix B) overlap only partially.

This paper describes it as “the smallest cross-city spread observed among UK types in this corpus” and offer no inferential claim. Possible interpretations include: violent-crime is well-fit by lag-and-rolling-mean features in all three UK cities, or violent-crime’s spatial structure is more similar across these three cities than other types’ structure. Replication on additional cities is required to distinguish these.

**Observation 2 (largest cross-UK PLL spreads: shoplifting, robbery, vehicle-crime).** Robbery and shoplifting exhibit the largest cross-city spreads in `nonzero_pll_mean`. Robbery in the UK literature concentrates at specific transport and entertainment nodes that vary between cities; shoplifting concentrates in cells that contain retail establishments, with high within-cell event clustering (multiple events per cell-month at a small number of cells, near-zero counts at the remainder) (Bowers, 2014; Brantingham & Brantingham, 1984; Eck, Clarke, and Guerette, 2007). The analysis tested whether shoplifting events concentrate in a small subset of *H3 cells*: yes, they do. The analysis did not test whether shoplifting events concentrate in a small subset of *establishments within H3 cells* — H3 r9 cells (~10.5 hectares) typically contain multiple establishments, so cell-level concentration is not direct evidence of facility-level concentration. The connection to Eck et al.’s “risky facilities” thesis is therefore at the level of consistency-with-pattern rather than direct test. The paper flags this distinction explicitly because the H3 spatial granularity does not adjudicate the facility-level claim.

**Observation 3 (calibration of enforcement-output categories).** Under the pipeline, drugs and possession-of-weapons calibration ratios in the UK cities differ from the typically-just-below-1.0 calibration of victimization-driven categories. Drugs and possession-of-weapons measure police activity (enforcement output), not latent victimization: recorded rates can drift in either direction with operational policing decisions the 14-feature stack does not see. Anti-social-behaviour, the second-largest UK category by event volume, sits between these two regimes — the data-recording category includes both responses to citizen complaints (victimization-like signal) and proactive police identification (enforcement-like signal), so this paper does not classify it as cleanly enforcement-driven or victimization-driven. The enforcement–victimization calibration distinction is reported as a candidate for future work, not as an established finding;  $n = 4$  cities provides insufficient power to support it inferentially.

### 4.3 Calibration improvement through iterative pipeline corrections (case study, not factorial decomposition)

Earlier configurations of this pipeline produced Boston `cal_ratio` of 0.849, approximately 0.07–0.08 below the three UK cities’ 0.92–0.93 in the same configurations. This paper describes below the sequential pipeline-development process by which this gap was closed to Boston `cal_ratio` = 0.964 in the configuration whose endpoint is reported as the canonical run; this is within rounding (0.001) of the Table 1 value of 0.963 — the 0.001 difference reflects subsequent rebuild-time

variation under the bootstrap-CI computation against the working-pipeline par-quet, not a substantive change. This paper explicitly characterizes this section as a chronological case study, not a factorial mechanism decomposition. The mechanisms below were applied sequentially, and this paper does not have the factorial  $2 \times 2 \times 2$  ablation that would isolate independent contributions. Numerical attributions to each mechanism (e.g., “+0.084”) are conditional on the order of application and should not be interpreted as additively decomposable contributions.

**Step 1: Time-decay weights, in sequence (cal\_ratio: 0.849  $\rightarrow$  0.867).**

Diagnostic analysis revealed that Boston’s mean test-period event rate (events per (cell, month, type)) was approximately 85 percent of its mean training-period event rate; the corresponding training-to-test rate ratios for the three UK cities were 88 to 95 percent, indicating smaller-magnitude drift than Boston but still some downward drift from training to test. Applying exponential time-decay weights at  $\lambda = 0.15$  — implemented for all four cities, not Boston only — corrected for this drift. UK city cal\_ratios moved correspondingly upward by approximately 0.01–0.02 from the pre-correction 0.92–0.93 baseline; Boston’s gain (+0.018) is comparable in magnitude to the UK gains rather than substantively larger, even though Boston’s training-to-test drift was the largest of the four cities. The implication is that time-decay weights at  $\lambda = 0.15$  partially correct for drift but do not eliminate it; the residual Boston gap relative to the UK cities is closed by the subsequent steps below, particularly the partial-bin correction that addresses a Boston-specific data-extraction issue.

**Step 2: Partial-bin exclusion, in sequence (cal\_ratio: 0.867  $\rightarrow$  0.951).**

As described in §3.4, the analysis revealed that Boston’s final test-set monthly bin contained only 10 days of data because the BPD extract was retrieved before the calendar month had ended. Adding the 28-day-coverage filter that excludes such bins moved Boston’s calibration\_ratio from 0.867 to 0.951, an attribution of +0.084 conditional on the time-decay weights from step 1 already being on. This paper describes this explicitly as the magnitude of an evaluation-time filtering correction in the pipeline, not a methodological insight; the 10-day partial bin was a consequence of when the extract was taken (i.e., the data was not yet complete for that calendar month), and the +0.084 attribution measures the impact of failing to filter that incomplete bin out of the test-set evaluation. Other research pipelines that already correctly handle partial bins should not expect a comparable gain from “applying partial-bin exclusion.”

**Step 3: Interleaved year-3 split, in sequence (cal\_ratio: 0.951  $\rightarrow$  0.964).**

The diagnostic also revealed that fractional 60/20/20 splits placed test windows disproportionately in winter (for the 2023–2026 windows). Replacing with the interleaved split described in §3.3 — under which validation and test bins alternate across the entire year-3 window so each block of seasonal or trend variation longer than two consecutive bins is straddled by both groups — eliminated this asymmetry, even though validation and test occupy disjoint calendar months under the alternating design.

**Null result on K5 vs. atomic taxonomy.** Before identifying the partial-bin issue, an initial hypothesis was that Boston’s K5 lumping was the cause of the calibration miss. The analysis tested this by running Boston on native NIBRS atomic categories at 5,000-event and 10,000-event thresholds (9 and 2 modeled types respectively). Median calibration ratios were 0.837 and 0.940 — not meaningfully different from the K5 `cal_ratio`. Whatever drove Boston’s pre-correction `cal_ratio` gap, K5 lumping was not it. This paper reports this as a useful null result documenting that an intuitive first guess about the source of the calibration miss was wrong.

**Caveat on attribution.** The +0.018, +0.084, and +0.013 attributions above are sequential, not factorial. The analysis has not run the full  $2 \times 2 \times 2$  cell of  $\{\text{decay} \times \text{partial-bin} \times \text{interleaved-split}\}$  on/off. The “+0.084 from partial-bin exclusion” is conditional on time-decay weights already being on; it might be larger or smaller in a from-scratch comparison. A factorial decomposition is left to future work. The case-study finding worth reporting is methodological: when a calibration discrepancy appears in cross-population evaluation, examine data-pipeline integrity (truncated bins, mismatched windows, sample composition) before changing model architecture or loss function. This recommendation does not require a factorial decomposition to motivate it.

#### 4.4 Four configuration-specific gradient-space negative results

During development the analysis tested four interventions targeting a divergence-detection diagnostic ( $\mu_{\text{max}} / \mu_{\text{median}} > 1000$ , where  $\mu_{\text{max}}$  and  $\mu_{\text{median}}$  are the maximum and median of the model’s predicted means  $\mu$  across the test set within a single (city, type) cell; defined ex ante for this project as a heuristic indicator of predictive blow-up risk, not adopted from prior literature) that fired on shoplifting in the three UK cities. As the paper explains below, the diagnostic itself proved to be a poor indicator — it conflated spatial concentration with predictive divergence — and the four interventions are reported as configuration-specific negative results rather than as categorical rejections.

**Tweedie loss at variance power  $p = 1.5$  (rejected in this configuration).**

The analysis replaced the Poisson training objective with Tweedie loss at variance power  $p = 1.5$ . Tweedie did not reduce shoplifting  $\mu_{\text{max}} / \mu_{\text{median}}$  ratios, and two Birmingham types (burglary and possession-of-weapons) that had been stable under Poisson exhibited training-time pathology: the validation loss ceased to improve within the first 1–8 boosting iterations (`best_iteration` between 1 and 8 across the two affected types), and the resulting test-set PLL collapsed to approximately  $-478$  per non-zero observation, indicating predictions effectively unrelated to the observed event rate. This paper does not generalize this as a rejection of Tweedie loss for count data; the paper reports only that  $p = 1.5$  was not viable in this specific pipeline and regime configuration. Other variance powers (notably  $p = 1.1$ ,  $p = 1.9$ ) were not tested. Practitioners considering Tweedie should not infer a categorical rejection from this study.

**Negative binomial at fixed dispersion (rejected for the configurations tested).** The analysis implemented a custom NB objective with dispersion parameter  $\alpha$  at fixed values  $\{0.1, 0.5, 1.0\}$ . At default learning rate, NB with  $\alpha \geq 0.5$  triggered `best_iter = 1` across every type in every city. Reducing the learning rate to 0.005 and 0.001 cleared this and converged the boosters, but the converged models over-predicted at the city-aggregate level: per-city `calibration_ratio` fell to the range 0.69–0.83 under NB at  $\alpha = 1.0$ ,  $lr = 0.005$ , well below the same-pipeline Poisson baseline (`cal_ratio` range 0.96–0.99 across the four cities, Table 1) in every city (i.e., the converged NB model’s mean exceeded the observed mean by approximately  $1.20\times$  to  $1.45\times$ ), while per-(city, type) `nonzero_pll_mean` improved (became less negative) by 6–14 percent in absolute magnitude across the (city, type) cells tested. The total event- weighted PLL was strictly worse than Poisson for every  $(\alpha, lr)$  combination tested.

**Important caveat.** This is *not* a rejection of the negative binomial likelihood family. Standard practice (Cameron & Trivedi, 2013; Hilbe, 2011) is to estimate  $\alpha$  from the data jointly with the mean — typically by maximum likelihood or empirical-Bayes — not to grid-search over fixed values. The test rejects “negative binomial loss with  $\alpha$  grid-searched at three values”; it does not test “negative binomial loss with  $\alpha$  estimated from the data.” NB-with-MLE- $\alpha$  might well succeed where NB-at-fixed- $\alpha$  fails, and the analysis did not test this. This is a substantial limitation of the negative result and the paper flags it explicitly.

**Training-target winsorization (mixed result; not a categorical rejection).** The analysis clipped training targets at the 99.9, 99.5, and 99.0 percentiles per (city, type). The 99.9-percentile threshold reduced shoplifting  $\mu$  ratios by 24–37 percent and did not degrade non-divergent types. By the paper’s metrics this is a net improvement at the 99.9 percentile. At 99.5 and 99.0 percentiles, however, shoplifting calibration overshot (`cal_ratio` 1.12–1.44, indicating clipping of real signal) and divergence migrated to adjacent types (London other-crime, Birmingham possession-of-weapons). The lesson is that clipping percentiles below 99.9 in this regime redistributes pathology rather than fixing it. The 99.9 percentile is potentially viable; the analysis did not adopt it because the analysis identified that the underlying divergence flag was measuring sparsity rather than predictive pathology (see “On the divergence flag itself,” below), so no clipping was needed.

**Single-type learning-rate reduction by factor of 5 (rejected for this specific configuration).** The analysis reduced the learning rate from 0.05 to 0.01 (a factor-of-5 reduction) for shoplifting only. With 500 maximum iterations at this rate, divergence flags cleared on all types, but shoplifting test PLL dropped catastrophically: from  $-0.38$  to between  $-31$  and  $-90$ , and `cal_ratio` collapsed to approximately 0.01 across the three UK cities (Leeds 0.017, Birmingham 0.012, London 0.013), which by the §3.5 convention (`cal_ratio` =  $\bar{y}/\bar{\mu}$ , where values below 1 indicate over-prediction) means the model’s predicted  $\mu$  exceeds the observed event rate by approximately 60- to 80-fold. The mechanism, on inspection, is that the reduced learning rate prevents the booster from learning the

cell-level zero-heavy spatial structure within the 500-iteration budget; predictions remain elevated relative to the actual spatial distribution of events, with  $\mu$  values orders of magnitude above  $\bar{y}$  on the many cells where shoplifting events are near-zero. The divergence flag, defined on  $\mu_{\text{max}} / \mu_{\text{median}}$ , is silenced not because  $\mu$  predictions become well-calibrated but because  $\mu$  becomes uniformly elevated (no outliers above the median because the median itself is high). The analysis did not test smaller reductions (e.g.,  $0.05 \rightarrow 0.025$ , factor of 2) or alternative methods such as reducing maximum iterations at the same learning rate. The result rejects the specific configuration tested, not learning-rate reduction in general.

**On the divergence flag itself.** The four interventions above were attempts to silence a divergence flag that, on inspection, was firing on a structural property of the data rather than a model failure. The ratio test  $\mu_{\text{max}} / \mu_{\text{median}} > 1000$  fires when median  $\mu$  approaches zero, which is structural for spatially-concentrated types (UK shoplifting zero rates 0.79–0.91 across the three UK cities; see §4.1). A predictive model that correctly identifies the small number of commercial cells where shoplifting concentrates will exhibit this ratio structurally. The diagnostic was therefore measuring sparsity rather than pathology — a finding that emerged from the four failed intervention attempts.

This paper reports a possible replacement absolute-scale gate. Specifically, two candidate forms were considered — a fixed-threshold form ( $\mu_{\text{max}} > 500$ ) and a data-relative form ( $\mu_{\text{max}} > 2 \times y_{\text{max}}$ ) — and adopted the data-relative form as the default replacement candidate, on the reasoning that it scales naturally with each (city, type) cell’s actual maximum observed count and so does not require setting an absolute threshold per regime. The authors are explicit that this replacement is **not validated** against any corpus that contains genuine divergence at those scales. In the corpus, the maximum predicted  $\mu$  across all (cell, month) observations within the three UK shoplifting (city, type) cells was about 70 (the largest single-cell-month predicted mean for shoplifting in any of London, Birmingham, or Leeds), well below the 500-fixed threshold; this maximum  $\mu$  was also below  $2 \times y_{\text{max}}$  for each of the three UK shoplifting (city, type) cells. The replacement gate is therefore inactive on the corpus regardless of which of the two forms is used. This paper reports the gate’s design as a candidate for future work, not as an evaluated diagnostic. Practitioners adopting the gate should validate it against their own corpora that include known divergent cases, and should choose between the fixed-threshold and data-relative forms based on whether their downstream use is comparing predictive divergence across regimes (data-relative preferred) or against an absolute upper-bound prediction tolerance (fixed-threshold preferred).

**Collective interpretation.** The four configuration-specific negative results share a pattern: when a diagnostic flag fires on a structural property of sparse count data (here, spatial concentration with high zero rate), interventions that bound the gradient’s response often shift the location of the flagged behavior across (city, type) cells rather than addressing the underlying property, or — as

in the lr-reduction case — degrade the predictions enough to clear the diagnostic threshold. In the specific configuration, the underlying issue was that the diagnostic flag was measuring the wrong property (spatial sparsity rather than predictive pathology). This paper does not generalize this pattern beyond the configurations tested.

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## 5. Discussion

### 5.1 What this paper establishes and what it does not

The paper establishes:

- **A methodological note (§4.1).** Aggregate event-weighted Poisson PLL is confounded by zero-rate composition. The confound is large enough to reshuffle the apparent ranking of cities depending on which evaluation metric is used in the four-city demonstration: three different metrics produce three different rank-1 cities, and no city holds rank across all three. `nonzero_pll_mean` and `calibration_ratio` are recommended as the primary cross-population pair, with `zero_rate` reported as descriptive context.
- **A negative-results catalog (§4.4).** In a specific empirical regime — heavy-tailed shoplifting concentration under monthly aggregation — four configuration-specific gradient-space interventions were tested against a divergence flag that on inspection was measuring spatial sparsity rather than predictive pathology. Three of the four (Tweedie loss at  $p=1.5$ , NB with grid-searched fixed  $\alpha$ , single-type lr-reduction) did not improve test predictions; the fourth (winsorization at the 99.9 percentile) was a net improvement that was not adopted because of the diagnostic’s mis-targeting. This paper documents all four explicitly so that practitioners working in similar regimes have a record.

The paper does not establish:

- **Architecture generalization across jurisdictions.** The four-city corpus is too small ( $n = 4$ ) and too institutionally homogeneous (all high-income OECD English-speaking common-law jurisdictions) to support a generalization claim. The narrow `calibration_ratio` band of  $[0.96, 0.99]$  across these four cities (Boston 0.963, Leeds 0.989, Birmingham 0.989, London 0.993) is consistent with both “the architecture generalizes” and “these four cities are too similar to distinguish the hypotheses.”
- **A principle about data hygiene vs. model tuning.** §4.3’s case study reports a sequential pipeline-development process; it does not establish a generalizable claim about the relative importance of data-pipeline integrity vs. model-architecture choices in count-model evaluation. The “+0.084 `cal_ratio` improvement” attributed to partial-bin exclusion is the magnitude of a specific extraction- freshness artifact in the pipeline (the BPD extract

was retrieved before the calendar month had ended; see §3.4), not a methodological lesson.

- **A factorial decomposition of pipeline mechanisms.** The three sequential mechanisms in §4.3 (time-decay weights, partial-bin exclusion, interleaved year-3 split) are not factorially independent; the attributed magnitudes are conditional on the order of application.
- **A categorical rejection of Tweedie, negative binomial, winsorization, or per-type learning-rate reduction.** The negative results are at fixed configurations and do not test the full hyperparameter or sub-class space of any of these techniques. NB-with-MLE- $\alpha$  in particular is untested.
- **A taxonomy non-generalization claim.** Appendix A reports that the existing UK Home Office 14-category taxonomy does not cluster stably into a smaller- $k$  partition under repeated clustering. This is a finding about the structure of the existing taxonomy, not about whether crime taxonomies generalize across jurisdictions.

## 5.2 Limitations

**Sample size.**  $n = 4$  cities. Quantitative inference in this paper is valid at the per-(city, type) cell level (36 cells, with bootstrap CIs); cross-city patterns across  $n = 4$  cities are descriptive, not inferential.

**Institutional homogeneity.** All four cities are in high-income OECD English-speaking common-law countries with similar policing institutions, similar data-recording traditions, and partially-overlapping regulatory environments. (City-level affluence and within-city deprivation profiles vary across the four cities; the homogeneity claim is at the country and policing-institution level, not the city or neighborhood level.) Generalization beyond this institutional context is unsupported.

**No external validation city in the primary design.** Every parameter choice in the pipeline was set using data from cities that also appear in the headline results:  $\lambda = 0.15$  (time-decay weight), the 28-day partial-bin threshold, and the K5 vs atomic-NIBRS taxonomy ablation were each set using Boston only; the 14-feature pruning was chosen using LightGBM split-importance gain on a London-trained model (§3.2). There is no held-out external-validation city for which the pipeline was frozen in the primary design. The four-city evaluation is therefore a within-sample fit, not a between-sample generalization test. The paper initially framed Appendix C.4 as a single post-hoc external-validation reassurance, but on inspection one of the three configuration choices examined in §C.4 (the 22→14 feature reduction) involves London-derived information; this paper therefore retracts that framing and now present §C.4 as a methodological limitation rather than a validation experiment. A properly-blinded external-validation study, in which the 22→14 feature reduction is re-derived using only Boston, Leeds, and Birmingham gain values before being applied to London, is left to follow-up work.



**Test-window context.** Test bins for all four cities fall in 2025–2026 (§2.1). Concurrent UK retail-crime dynamics over the test window — including a widely-reported cost-of-living-driven rise in shoplifting volumes — are not independently verified or modeled in this paper; results are conditional on the specific test window. §C.3 reports a sensitivity analysis under an alternative test-window split (Window A: year-3 first half as validation, year-3 second half as test). The structural three-way reshuffle finding from §4.1 holds under Window A; however, the rank-1 city on `calibration_ratio` shifts from London (canonical) to Leeds (Window A), driven by a London-specific `calibration_ratio` drop of 0.029 that exceeds bootstrap-CI overlap with the canonical estimate while the other three cities’ shifts stay within CI overlap. Practitioners using `calibration_ratio` for cross-jurisdictional ranking decisions when the inter-city margin is narrow (under 0.005, as between Leeds, Birmingham, and London under the canonical split) should report bootstrap CIs and at least one alternative-test-window sensitivity rather than treating the point-estimate ordering as robust.

**Spatial granularity for risky-facilities claims.** H3 r9 cells contain multiple establishments. Cell-level concentration findings are at best consistent with, but do not directly test, facility-level concentration patterns. Direct facility-level testing requires point-of-interest (POI) density features that this paper does not include.

**Single-objective evaluation.** The analysis evaluates exclusively on Poisson predictive log-likelihood and calibration ratio. Alternative evaluation targets (interval coverage, tail accuracy, ranked-area concentration curves, prediction-area indices) might produce different orderings; the recommendations are within the PLL/calibration framework and do not transfer mechanically.

**Boston K5 to UK mapping.** No canonical NIBRS-to-Home-Office mapping exists; the heuristic mapping (§2.4) is documented but not validated. Per-type Boston-UK comparisons in this paper should be read as approximate.

**Negative-results scope.** The four negative results in §4.4 are at fixed configurations. This paper does not claim they generalize beyond the tested regime or beyond the specific hyperparameter settings tested. Practitioners working in different regimes should not infer that these techniques will fail in their settings.

**Broader critiques of predictive policing.** A substantial literature documents potential harms from predictive-policing systems built on historical police data, including feedback loops, racial-bias amplification, and disparate impact (Lum & Isaac, 2016; Richardson, Schultz, and Crawford, 2019; Heaven, 2020). The methodological note in this paper — that aggregate event-weighted Poisson PLL is confounded by zero-rate composition — is upstream of those concerns: it is a measurement issue about how performance is reported, not a recommendation that any particular city deploy any particular system. Whether a given deployment is appropriate depends on factors beyond predictive performance, including data-generation processes, feedback mechanisms, and oversight structures, none of which this paper addresses.

### 5.3 Recommendations for cross-population count-model evaluation

Based on the methodological note in §4.1, this paper offers the following recommendations:

1. Report `nonzero_pll_mean` and `calibration_ratio` jointly, with `zero_rate` as context, when comparing count-model performance across populations.
2. Report bootstrap or analytic confidence intervals on these per-cell metrics. Cross-population comparisons should accompany point estimates with the corresponding interval estimates (or, equivalently, with per-cell sample sizes that bound how informative the point estimate can be).
3. State the aggregation function used for any “calibration” metric explicitly. Per-row  $\text{mean}(y_i/\mu_i \mid y_i > 0)$  and aggregate  $\text{mean}(y)/\text{mean}(\mu)$  are conceptually distinct and can disagree on the same data; this paper uses the aggregate form throughout, as discussed in §4.1.
4. When event-weighted aggregate PLL is reported, report the composition of zero rates and central-tendency non-zero  $\mu$  values (e.g., median or mean predicted  $\mu$  over the subset of test observations with  $y_i > 0$ ) across the populations being compared, so that readers can detect scale-confound effects.
5. When a calibration discrepancy emerges in cross-population evaluation, investigate data-pipeline integrity and evaluation-window symmetry before changing model architecture or loss function. The authors arrived at this procedural ordering through the case study in §4.3 — where the partial-bin and time-decay corrections together accounted for the `cal_ratio` gap — rather than from a deliberately designed comparison. This paper recommends the ordering as a procedural heuristic that would have saved the four-intervention exploration documented in §4.4, not as an established principle.

For practitioners considering interventions targeting heavy-tailed spatially-concentrated count data, the §4.4 catalog provides four specific configurations that did not work in the configuration regime. This paper recommends that such practitioners (a) test against ground-truth-known divergent cases, not against ratio-based diagnostic flags that may instead be sparsity-driven; (b) include NB-with-MLE- $\alpha$  among the tested loss-family alternatives, since the null result is on NB-at-fixed- $\alpha$  only; and (c) test winsorization at the 99.9 percentile (which was viable in the configuration regime) before testing more aggressive thresholds.

### 5.4 Future work

A natural extension is a cross-jurisdiction validation with a substantively larger sample with held-out external cities, a factorial decomposition of pipeline-correction mechanisms, and inclusion of cities outside the high-income OECD English-speaking common-law institutional context. Direct facility-level testing

of risky-facility hypotheses would require integration of POI density features beyond the feature stack used here. A negative-results catalog with broader hyperparameter coverage (Tweedie at multiple variance powers, NB-with-MLE- $\alpha$ , winsorization at finer percentile granularity) would strengthen the configuration-specific findings reported in §4.4. Sub-monthly UK data, if accessible through force-specific data-sharing agreements, would enable direct comparison of feature-set richness vs. granularity effects in absolute predictive quality.

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## 6. Conclusion

This paper documents that aggregate event-weighted Poisson predictive log-likelihood, a commonly-reported metric in cross-population count-model evaluation, is confounded by zero-rate composition. The confound is large enough to reshuffle the apparent ranking of cities depending on which evaluation metric is used: in the four-city demonstration, three different metrics produce three different rank-1 cities — Birmingham on event-weighted PLL, Boston on `nonzero_pll_mean`, London on `calibration_ratio` (the latter under the canonical split; the rank-1 city on `calibration_ratio` is window-fragile, shifting to Leeds under an alternative-test-window sensitivity reported in §C.3) — and no city holds rank across all three. This paper proposes `nonzero_pll_mean` and `calibration_ratio` as the primary cross-population pair, with `zero_rate` reported as descriptive context, and this paper recommends joint reporting of all three with bootstrap confidence intervals.

The analysis additionally document four configuration-specific gradient-space interventions — Tweedie loss at  $p = 1.5$ , negative binomial loss at three fixed dispersion settings, training-target winsorization at three percentiles, single-type learning-rate reduction by factor of 5 — evaluated against a divergence flag that, on inspection during those interventions, was firing on spatial sparsity rather than predictive pathology. Three of the four (Tweedie, NB, single-type lr-reduction) did not improve predictions in the specific empirical regime; the fourth (winsorization at the 99.9 percentile) was a net improvement that was not adopted because of the diagnostic’s mis-targeting. The authors are explicit that these findings are configuration-specific, not categorical rejections of the techniques involved, and the analysis identifies NB-with-MLE- $\alpha$  (negative binomial with  $\alpha$  estimated from the data by maximum likelihood or empirical Bayes) as a notable untested alternative that future work should evaluate.

The four-city LightGBM application is presented as a demonstration of the methodological note and as the empirical context in which the §4.4 configuration-specific negative results were observed (each applied to specific (city, type) cells — principally shoplifting in the three UK cities — not across all 36 (city, type) cells in the corpus). This paper does not claim cross-jurisdictional generalization for the architecture; the four-city sample is too small and too institutionally homogeneous to support such a claim. The decomposition of Boston’s calibra-

tion improvement is presented as a chronological case study, not a factorial decomposition; sequential mechanism attributions are conditional on the order of application.

Two findings are the most likely to transfer beyond this study: the §4.1 scale-confound observation, which is algebraic and applies to any Poisson count model evaluated across populations of differing zero-rate composition; and the §5.3 procedural recommendation (recommendation 5) to investigate data-pipeline integrity and evaluation-window symmetry before changing model architecture or loss function when cross-population calibration discrepancies appear.

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## Appendix A. Taxonomy investigation: structure of the existing UK Home Office taxonomy

This appendix reports a subsidiary investigation into whether the existing UK Home Office 14-category taxonomy clusters stably into a smaller- $k$  partition under repeated clustering. The authors are explicit that the finding is about the **structure of an existing taxonomy**, not about whether crime taxonomies generalize across jurisdictions. The clustering analysis takes the 14 pre-defined categories as input; what it can find is whether those 14 categories cluster stably into fewer groups, not whether different jurisdictions’ taxonomies are commensurable. The latter is a stronger claim that this study does not test.

### A.1 Clustering input

The clustering analysis uses two distinct feature families (constructed from the same event data via different aggregation choices), which the appendix refers to throughout the appendix as Track 1 and Track 2. **Track 1 (scale-invariant features)** comprises seven per-category summary features computed per city for each of the three UK cities: `log_volume` ( $\log_{10}(\text{events} + 1)$ ), `hour_entropy` (Shannon entropy in bits, i.e., log base 2, over the 24-hour event-time distribution), `weekend_ratio` ((Saturday + Sunday events) / total), `seasonal_amplitude` (mean monthly event count over summer months May–August divided by mean monthly event count over winter months November–February), `spatial_gini` (Gini coefficient over per-cell event counts within the city’s H3 cell set as defined in §2.3, computed for the focal category), `spatial_spread` ( $\log_{10}(\text{unique cells} + 1) / \log\_volume$ ), and `peak_hour_normalized` (the hour-of-day index 0–23 at which the focal category’s hourly event count is maximized, divided by 23 to rescale into  $[0, 1]$ ). **Track 2 (co-occurrence features)** is the  $14 \times 14$  pairwise Spearman correlation matrix of weekly category counts computed per city across that city’s (cell, week) pairs; this matrix is used directly as the inter-category distance signal under the metric  $d = 1 - \rho$ . The two tracks differ in what they represent: Track 1 summarizes each category’s marginal temporal-spatial profile, while Track 2 captures which categories tend to co-occur in the same cells and weeks. Strong agreement between the two tracks would indicate that categories

with similar marginal profiles also co-occur; weak agreement would indicate the two signals carry independent information about category structure.

## A.2 Clustering methods

The analysis applied three clustering algorithms — Ward hierarchical linkage, spectral clustering on a similarity-graph Laplacian (Gaussian affinity constructed from Track 1 feature distances; the Track 2 Spearman matrix used directly as similarity), and  $k$ -medoids — at each of  $k \in \{3, 4, 5, 6, 7\}$ . The analysis evaluated cluster quality using the silhouette score (Rousseeuw, 1987) and the gap statistic (Tibshirani, Walther, and Hastie 2001), and assessed cluster stability by bootstrap resampling (50 resamples).

## A.3 Sensitivity design

The analysis ran 36 configurations per city in the sensitivity sweep, defined by the factorial product of feature-set choices, distance metrics, standardization options, and volume-weighting options after excluding degenerate combinations (e.g., feature-set / distance-metric pairings on which the metric is undefined). Across the three UK cities, this yields 108 configurations total. The complete configuration enumeration with each factor’s levels is in the project repository.

## A.4 Findings

- Track 1 (scale-invariant features) vs. Track 2 (co-occurrence) ARI per city at  $k = 5$ : London 0.118, Birmingham 0.056, Leeds 0.084 — all well below the  $\text{ARI} \geq 0.5$  rule-of-thumb threshold for moderate agreement between two partitions of the same items, so the two feature families produce different partitions of the same 14 categories rather than recovering a shared latent structure.
- Cross-city ARI at  $k = 5$  (chosen because the gap-optimal  $k = 3$  partition is degenerate, see fourth bullet below;  $k = 5$  gives a non-degenerate partition for all three cities): London  $\leftrightarrow$  Birmingham =  $-0.039$ , London  $\leftrightarrow$  Leeds =  $-0.077$ , Birmingham  $\leftrightarrow$  Leeds =  $+0.039$ . Two of three cross-city values are negative; an ARI of zero is the chance-baseline expected value under random partitions (Hubert & Arabie, 1985), so values near zero indicate no systematic agreement and slightly-negative values indicate slightly-less-than-chance agreement (within sampling uncertainty given the 14-category partition size;  $\binom{14}{2} = 91$  within-partition category pairs are the basis for ARI computation). The three UK cities produce different partitions of the same 14-category schema even when feature family and  $k$  are held fixed.
- Gap-statistic consensus  $k$ : across 18 (city  $\times$  feature-family) combinations — namely, the 3 UK cities crossed with 6 feature-family configurations drawn from the sensitivity sweep (subsets and combinations of the seven scale-invariant features defined in §A.1, plus the co-occurrence-based representation, with the precise enumeration given in the project repository) — all 18 returned  $k = 3$  as the gap-optimal cluster count.

- Resulting  $k = 3$  partition: degenerate 10-2-2 (ten of the 14 categories assigned to one cluster, two each to the other two clusters) — gap-optimal but assigning the bulk of the taxonomy to a single cluster, which provides no actionable subdivision of the 14 categories.
- Across the 108 sensitivity configurations:  $k = 3$  was selected as the gap-optimal count in 70.4 percent (76/108); among these 76 selections, the mean pairwise ARI between (city, config) partitions was 0.150 — well below the  $\text{ARI} \geq 0.5$  rule-of-thumb threshold for moderate agreement. This paper uses the chance-corrected ARI of Hubert and Arabie (1985), under which the expected value of ARI for two random partitions is zero.

### A.5 Interpretation

The existing UK Home Office 14-category taxonomy does not cluster stably into a smaller- $k$  partition under any of the clustering configurations tested. This is consistent with the hypothesis that the 14 categories were defined for operational discriminability, not to reflect a recoverable smaller- $k$  latent structure. This paper does not interpret this as evidence about cross-jurisdiction taxonomy generalization, which would require a different study design (e.g., clustering on event-level features rather than category-summary features; comparison of fine-grained category boundaries across jurisdictions).

The analysis computed the Boston K5 to UK ARI under the heuristic mapping documented in §2.4 (mean  $\text{ARI} = 0.109$  across the 108 configurations). This is the level of agreement between *the chosen mapping* and the empirical UK clustering — a small positive value indicating poor agreement, well below the  $\text{ARI} \geq 0.5$  moderate-agreement rule-of-thumb cited in §A.4 — not a substantive comparison of Boston’s and UK’s underlying taxonomies. This paper reports it for completeness but do not draw inferential conclusions from it.

A parallel finding in a different methodological domain comes from McCarthy et al. (2025), who apply factor analysis to UCR/NIBRS at the individual-offender level and find that the property and societal factor structures replicate cleanly across their two justice-involved samples while the violent-crime factor structure does not — that is, the conventional violent-vs-non-violent partition is not cleanly recoverable in their data even though the broader UCR/NIBRS system is supported. The category-profile clustering finding (using marginal spatial and temporal features per category) is methodologically distinct from their factor-analytic finding, but the two together are consistent with the broader observation that conventional crime-category aggregations are not fully recoverable from data.

## Appendix B. Per-(city, type) decomposition with bootstrap confidence intervals

**Table B.1.** Per-(city, type) decomposition with 95 percent bootstrap confidence intervals (1,000 resamples of test (cell, month) observations within each (city, type) cell). All values reported here are from the canonical (year-3 calendar) test split described in §3.3; the Window A sensitivity (§C.3) decomposes the same metrics under the year-3 half-window split. The `dispersion_index` column reports the empirical variance-to-mean ratio of the test-set observed counts within each (city, type) cell ( $\text{var}(y) / \text{mean}(y)$  over the (cell, month) observations); a value of 1.0 indicates a Poisson-consistent distribution, values  $> 1$  indicate overdispersion. The `mean(y|y>0)` column reports the arithmetic mean of  $y$  over the subset of test observations with  $y > 0$  within each cell. These two columns describe the structure of the observed test data and are not model outputs; they support interpretation of the `nonzero_pll_mean` and `calibration_ratio` values to their left.

| City   | Type                  | n_test_events | calibration_ratio     | nonzero_pll_mean        | mean(y y>0) | dispersion_index | mean(y y>0) |
|--------|-----------------------|---------------|-----------------------|-------------------------|-------------|------------------|-------------|
| Boston | band_1                | 6,609         | 0.942<br>(0.91, 0.97) | -2.14<br>(-2.21, -2.07) | 0.74        | 15.968           | 3.53        |
| Boston | band_4                | 4,262         | 0.903<br>(0.87, 0.94) | -1.81<br>(-1.86, -1.77) | 0.73        | 4.863            | 2.20        |
| Boston | band_2                | 4,155         | 1.040<br>(1.00, 1.08) | -1.98<br>(-2.06, -1.92) | 0.75        | 13.141           | 2.31        |
| Boston | band_0                | 3,410         | 1.002<br>(0.96, 1.04) | -1.95<br>(-2.00, -1.90) | 0.74        | 2.959            | 1.81        |
| Leeds  | violent-crime         | 14,900        | 1.026<br>(1.01, 1.05) | -2.11<br>(-2.15, -2.07) | 0.38        | 6.923            | 3.66        |
| Leeds  | shoplifting           | 3,665         | 0.959<br>(0.90, 1.02) | -3.08<br>(-3.40, -2.81) | 0.90        | 21.453           | 5.81        |
| Leeds  | anti-social-behaviour | 3,228         | 1.117<br>(1.06, 1.17) | -2.17<br>(-2.26, -2.09) | 0.73        | 3.524            | 1.83        |
| Leeds  | public-order          | 3,067         | 0.904<br>(0.87, 0.94) | -1.84<br>(-1.89, -1.79) | 0.74        | 3.616            | 1.79        |
| Leeds  | other-theft           | 2,674         | 0.986<br>(0.94, 1.03) | -1.94<br>(-1.99, -1.89) | 0.76        | 3.477            | 1.70        |

| City       | Type                          | n_test_events | calibration              | nonzero                    | mean | dispersion | mean(y<br>index0) |
|------------|-------------------------------|---------------|--------------------------|----------------------------|------|------------|-------------------|
| Leeds      | criminal-<br>damage-<br>arson | 2,521         | 0.896<br>(0.86,<br>0.93) | -1.85<br>(-1.90,<br>-1.79) | 0.75 | 1.815      | 1.56              |
| Leeds      | vehicle-<br>crime             | 2,169         | 0.909<br>(0.87,<br>0.95) | -2.01<br>(-2.08,<br>-1.95) | 0.78 | 1.795      | 1.49              |
| Leeds      | burglary                      | 2,020         | 0.973<br>(0.92,<br>1.02) | -2.04<br>(-2.11,<br>-1.97) | 0.78 | 1.634      | 1.41              |
| Birmingham | violent-<br>crime             | 31,902        | 1.021<br>(1.00,<br>1.04) | -2.11<br>(-2.16,<br>-2.07) | 0.36 | 5.068      | 3.41              |
| Birmingham | shoplifting                   | 6,942         | 1.033<br>(0.99,<br>1.08) | -2.96<br>(-3.11,<br>-2.81) | 0.91 | 20.351     | 5.27              |
| Birmingham | vehicle-<br>crime             | 6,380         | 0.964<br>(0.94,<br>0.99) | -1.79<br>(-1.83,<br>-1.76) | 0.70 | 1.620      | 1.45              |
| Birmingham | criminal-<br>damage-<br>arson | 5,662         | 1.019<br>(0.99,<br>1.05) | -1.90<br>(-1.94,<br>-1.86) | 0.74 | 1.724      | 1.48              |
| Birmingham | other-<br>theft               | 4,721         | 0.988<br>(0.96,<br>1.02) | -1.98<br>(-2.02,<br>-1.94) | 0.79 | 2.526      | 1.56              |
| Birmingham | anti-<br>social-<br>behaviour | 4,719         | 0.825<br>(0.80,<br>0.85) | -1.88<br>(-1.92,<br>-1.84) | 0.80 | 2.294      | 1.57              |
| Birmingham | public-<br>order              | 4,276         | 0.899<br>(0.87,<br>0.93) | -1.96<br>(-2.01,<br>-1.92) | 0.81 | 2.568      | 1.57              |
| Birmingham | burglary                      | 3,571         | 0.898<br>(0.86,<br>0.93) | -1.98<br>(-2.02,<br>-1.94) | 0.81 | 1.438      | 1.29              |
| Birmingham | drugs                         | 2,584         | 1.094<br>(1.04,<br>1.15) | -2.56<br>(-2.63,<br>-2.49) | 0.89 | 3.127      | 1.59              |
| Birmingham | robbery                       | 1,987         | 0.977<br>(0.93,<br>1.02) | -2.31<br>(-2.37,<br>-2.25) | 0.90 | 1.877      | 1.38              |
| Birmingham | possession-<br>of-<br>weapons | 1,806         | 1.128<br>(1.07,<br>1.19) | -2.52<br>(-2.59,<br>-2.45) | 0.90 | 1.430      | 1.25              |



| City   | Type                  | n_test_events | calibration<br>mean      | nonzero<br>mean            | permeate | dispersion_index | mean(y<br>x=0) |
|--------|-----------------------|---------------|--------------------------|----------------------------|----------|------------------|----------------|
| London | violent-crime         | 57,597        | 1.037<br>(1.03,<br>1.05) | -2.13<br>(-2.15,<br>-2.11) | 0.23     | 5.185            | 4.17           |
| London | anti-social-behaviour | 53,116        | 0.999<br>(0.99,<br>1.01) | -2.21<br>(-2.23,<br>-2.18) | 0.30     | 5.376            | 4.21           |
| London | theft-from-the-person | 31,844        | 0.927<br>(0.91,<br>0.95) | -2.18<br>(-2.22,<br>-2.14) | 0.60     | 31.898           | 4.37           |
| London | other-theft           | 30,863        | 1.005<br>(0.99,<br>1.02) | -1.83<br>(-1.86,<br>-1.81) | 0.47     | 12.673           | 3.25           |
| London | shoplifting           | 22,388        | 1.064<br>(1.03,<br>1.09) | -2.93<br>(-3.04,<br>-2.84) | 0.79     | 18.211           | 5.84           |
| London | vehicle-crime         | 15,178        | 0.883<br>(0.87,<br>0.90) | -1.61<br>(-1.63,<br>-1.60) | 0.52     | 1.898            | 1.77           |
| London | public-order          | 13,807        | 1.013<br>(0.99,<br>1.03) | -1.90<br>(-1.93,<br>-1.87) | 0.62     | 2.715            | 2.00           |
| London | drugs                 | 12,691        | 1.146<br>(1.11,<br>1.19) | -2.44<br>(-2.62,<br>-2.30) | 0.67     | 6.114            | 2.14           |
| London | burglary              | 11,422        | 0.919<br>(0.90,<br>0.94) | -1.72<br>(-1.75,<br>-1.70) | 0.62     | 1.776            | 1.69           |
| London | criminal-damage-arson | 10,887        | 0.955<br>(0.94,<br>0.98) | -1.70<br>(-1.73,<br>-1.67) | 0.62     | 1.654            | 1.59           |
| London | robbery               | 8,662         | 0.922<br>(0.90,<br>0.94) | -1.85<br>(-1.89,<br>-1.82) | 0.73     | 3.292            | 1.80           |
| London | bicycle-theft         | 4,597         | 0.903<br>(0.88,<br>0.93) | -1.89<br>(-1.93,<br>-1.85) | 0.82     | 1.733            | 1.41           |
| London | other-crime           | 2,341         | 1.051<br>(1.01,<br>1.10) | -2.41<br>(-2.46,<br>-2.36) | 0.89     | 1.298            | 1.17           |

**Table B.2.** Per-city summaries with equal weight per type (rather than the

event-weighting used in Table 1), under the canonical (year-3 calendar) test split described in §3.3. The type-equal-weighted aggregator is the arithmetic mean across types of the corresponding per-(city, type) value in Table B.1: for **nonzero\_pll\_mean** this is the mean of per-type means; for **calibration\_ratio** this is the mean of per-type ratios (i.e., a mean-of-ratios rather than a ratio of equal-weighted sums); for **agg\_pll** this is the mean of per-type aggregate PLL values. Bootstrap CIs are computed from the same observation-level resamples as Table B.1, propagated through the same type-equal-weighted aggregator. The **agg\_pll** values match Table 1 because the number of test (cell, month) rows is equal across types within each city in this corpus (every modeled type has the same complete cell  $\times$  month grid as its row set, a property of the test-window construction). Note that this row-count equality is distinct from event-count equality: the **n\_test\_events** column in Table B.1 (which counts  $\sum y$  per type, not row count) varies sharply across types within a city. The **agg\_pll** match is a row-count property; under non-equal per-type row counts the type-equal and event-weighted aggregations of **agg\_pll** would differ. The **nonzero\_pll\_mean** and **calibration\_ratio** columns differ from Table 1 because their effective per-type weights (count of nonzero observations and  $\Sigma\mu$  respectively) vary across types regardless of total observation count.

| City       | n_types | agg_pll<br>(type-equal-<br>weighted) | nonzero_pll_mean<br>(type-equal-<br>weighted) | calibration_ratio<br>(type-equal-<br>weighted) |
|------------|---------|--------------------------------------|-----------------------------------------------|------------------------------------------------|
| Boston     | 4       | -0.69 (-0.71,<br>-0.68)              | -1.97 (-2.00,<br>-1.94)                       | 0.972 (0.95,<br>0.99)                          |
| Leeds      | 8       | -0.76 (-0.77,<br>-0.75)              | -2.13 (-2.18,<br>-2.09)                       | 0.971 (0.95,<br>0.99)                          |
| Birmingham | 11      | -0.62 (-0.62,<br>-0.61)              | -2.18 (-2.19,<br>-2.16)                       | 0.986 (0.98,<br>1.00)                          |
| London     | 13      | -1.04 (-1.04,<br>-1.03)              | -2.06 (-2.08,<br>-2.05)                       | 0.986 (0.98,<br>0.99)                          |

## Appendix C. Reproducibility details and sensitivity analyses

### C.1 Compute environment

Linux server; 20 cores; 62 GB RAM. PostgreSQL 14 with PostGIS 3.4. Python 3.11; H3-py 4.4.2; LightGBM  $\geq 4.0$  (precise version recorded with each Git SHA in **experiments/** per §C.2); scikit-learn, pandas, and numpy versions are pinned in the project **requirements.txt**. Random seeds (consistent with §3.6): numpy random.Generator initialized via `np.random.default_rng(seed=42)` for all dataset-construction randomness; seeds 1 through 5 passed to `lightgbm.train()` for booster initialization and sub-sampling across the five Window A ensemble members for all four cities, and across the five canonical Phase 2 ensemble

members for the three UK cities. Boston canonical Phase 2 predictions are from a single-seed v15b run (LightGBM seed = 42) retained via symlink, as documented in §3.6.

## C.2 Pipeline modules and Git SHAs

Configuration files, Git SHAs, and seeds for each reported training run will be released in the public methodological-subset repository (see §3.6) under the `experiments/` directory. The components below are in the public-release scope:

- Taxonomy investigation (Appendix A clustering analysis): primary analysis script and 108-configuration sensitivity script.
- Per-city LightGBM training and evaluation: production script.
- Per-(city, type) decomposition with bootstrap CIs: corrected-metrics rebuild script.
- Sensitivity analyses (§C.3 and §C.4 below): alternative-test-window and external-validation scripts.

## C.3 Test-window sensitivity (Window A)

To test whether the §4.1 finding depends on the §3.3 interleaved year-3 split, the analysis re-ran the four-city demonstration under an alternative test-window split, denoted *Window A*: a year-aligned half-window split in which the validation set is year-3 first half and the test set is year-3 second half. Under this split the UK cities have 6 validation and 6 test bins each (matching the §3.3 baseline); Boston has 8 validation and 7 test bins (versus 7 / 8 in §3.3, the asymmetry arising from the different parities of contiguous-half vs interleaved partitions of the 15 year-3 bins remaining after partial- bin exclusion). Window A is otherwise identical to §3.3: identical training-bin set, identical 28-day partial-bin filter, identical  $\lambda = 0.15$  time-decay weights, identical 14-feature stack, identical hyperparameters. For all four cities under Window A, five seeds (1, 2, 3, 4, 5) were run with predictions averaged across seeds; per-city metrics were evaluated on the 5-seed ensemble with bootstrap CIs from 1,000 observation-resamples. Bootstrap CIs reflect observation-sampling uncertainty around the ensemble point estimate and not inter-seed prediction variability; inter-seed prediction variability is much smaller than observation-sampling uncertainty for the metrics reported here. Note that this differs from the canonical Phase 2 construction, where the three UK cities use a 5-seed v15f ensemble but Boston uses a single-seed v15b output (see §3.6). The Boston canonical→Window-A comparison reported below is therefore confounded by pipeline configuration (single-seed v15b vs 5-seed v15f ensemble) in addition to test-window choice; the paper reports Boston’s Window A delta for completeness but interpret it as a multi-source shift rather than a clean window-only sensitivity. The three UK cities’ deltas are clean window-only sensitivities, holding pipeline configuration constant.

**Structural finding preserved.** The §4.1 three-way reshuffle holds under Window A: three different cities take rank-1 across the three recommended

metrics under both windows, and no city holds rank across all three under either. Birmingham is rank-1 on event-weighted aggregate PLL under both (canonical  $-0.615$ ; Window A  $-0.596$ ). Boston is rank-1 on `nonzero_pll_mean` under both (canonical  $-1.97$ ; Window A  $-1.94$ ), though this rank assignment for Boston should be read with the caveat from §3.6 that canonical Boston is single-seed v15b while Window A Boston is 5-seed v15f; the rank-1 status under both windows is suggestive of stability but not a clean like-for-like comparison.

**Calibration\_ratio rank-1 is window-dependent.** The rank-1 city on `calibration_ratio` shifts from London under the canonical split (point estimate 0.993) to Leeds under Window A (point estimate 0.977). Per-city `calibration_ratio` shifts under Window A relative to the canonical split:

- Boston:  $0.963 \rightarrow 0.959$  ( $\Delta = -0.004$ )
- Leeds:  $0.989 \rightarrow 0.977$  ( $\Delta = -0.012$ )
- Birmingham:  $0.989 \rightarrow 0.976$  ( $\Delta = -0.013$ )
- London:  $0.993 \rightarrow 0.965$  ( $\Delta = -0.029$ ; computed from underlying 4-decimal values 0.9932 and 0.9645, which round to 0.029 at 3 dp; subtracting the displayed 3-dp values would suggest  $-0.028$ )

Three of the four shifts (Boston, Leeds, Birmingham) sit within bootstrap-CI overlap of the canonical estimate; only London’s shift exceeds CI overlap (canonical 95% CI  $[0.988, 0.999]$  vs Window A 95% CI  $[0.959, 0.970]$ , completely disjoint). The cross-window rank-1 swap on `calibration_ratio` is therefore driven by London’s specific window-dependence rather than by Leeds gaining ground; under either window, Leeds and Birmingham point estimates are within 0.001 of each other (canonical: tied at 0.989 to three decimal places; Window A: 0.977 vs 0.976), and within Window A all six pairwise inter-city `calibration_ratio` CIs overlap. The within-Window-A cross-city ordering is therefore not statistically distinguishable on `calibration_ratio`; the rank-1 swap is a between-window observation driven by a single city.

**Aggregate PLL is window-stable.** Per-city event-weighted aggregate PLL deltas across the four cities range from  $+0.000$  (Boston) to  $+0.027$  (Leeds), all in the direction of less-negative PLL under Window A (i.e., all four cities have slightly better-looking PLL on the second-half test window than on the interleaved-bin test set), and all well below the 0.10 PLL sanity threshold the analysis set in advance for assessing window-induced shifts.

**Calibration uniformity under Window A.** Under Window A no city’s `calibration_ratio` bootstrap CI contains 1.0; all four CIs sit below 1.0, indicating mild over-prediction at the city-aggregate level uniformly across cities. Under the canonical split, only Leeds has a `calibration_ratio` CI whose upper bound reaches 1.0 at 3-decimal-place precision (upper = 1.003); Boston (0.981), Birmingham (0.999), and London (0.999) all have upper bounds strictly below 1.0. Window A surfaces over-prediction more sharply than the canonical split. The pattern is consistent with two non-exclusive readings: a property of the year-3 second-half test window specifically, or systematic mild over-prediction

that the canonical interleaved split happens to attenuate at narrow margins. Distinguishing them is not possible from a single window pair.

**Implication for the §4.1 argument.** This sensitivity strengthens rather than weakens the §4.1 methodological argument. The structural finding (three-way reshuffle, no city holds rank across all three) is robust to the test-window choice; the specific identity of the rank-1 city on `calibration_ratio` is window-fragile when the inter-city margin is small (under 0.005 between the rank-1 and rank-2 cities under canonical) relative to per-city window-induced shifts (up to 0.029 for London). This paper reports this as direct evidence that aggregate `calibration_ratio` orderings should not be over-interpreted at narrow margins, consistent with the paper’s broader argument against single-metric cross-population comparisons.

Detailed per-(city, type) Window A bootstrap CIs are in the project repository at `paper_tables_v15f_window_A/`.

#### **C.4 Methodological limitation: configuration choices are not strictly external-to-London**

The pipeline whose results the paper reports for the three UK cities is internally named `v15f_ensemble`; `v8` through `v15f` are intermediate development versions, with `v15f` being the final reported configuration and the `_ensemble` suffix referring to its 5-seed ensemble structure (§3.6). Boston canonical Phase 2 results are from the `v15b` single-seed predecessor configuration retained via symlink, as documented in §3.6. Its configuration choices include three layered decisions: (a) the 22→14 feature reduction at `v15b` (the 22-feature predecessor and the eight dropped features are enumerated in §3.2), which dropped features with less than 2 percent split-importance gain in a London LightGBM model; (b) the 28-day partial-bin exclusion threshold introduced at `v8` in response to a Boston-specific data-boundary artifact (a 10-day truncated final month) and applied uniformly to all daily-resolution cities; (c) the interleaved year-3 split design, applied uniformly across cities by the formula  $(b-24) \bmod 2$ . Of these, decision (a) was driven by London-specific feature-importance metrics, so the `v15f_ensemble` configuration as deployed is not strictly external-to-London. Decisions (b) and (c) are city-blind by construction. Downstream of the 22→14 step, the 14-feature stack is applied to London with no London-specific hyperparameters, no London-specific feature engineering, and no London-specific ensembling weights — the per-city training is uniform across the four cities. This paper therefore retracts the original §C.4 external-validation framing and present it as a methodological limitation: a properly-blinded external-validation experiment, in which the 22→14 step is re-derived using only Boston, Leeds, and Birmingham gain values before being applied to London, is left to a follow-up paper. The London row of Table 1 is reported as in-sample performance under the as-deployed configuration.

### C.5 Known pipeline issues resolved during investigation

- PostgreSQL default-database-name mismatch (resolved at initial environment setup).
- PostgreSQL peer-authentication user mismatch (resolved at initial environment setup).
- H3-py v4 API change: `grid_disk` returns list, not set (resolved at first run against H3-py v4).
- UK parquet files contained both a pre-bucketed `crime_type` column and the raw `original_category` column; the latter (14 native categories) was used.
- UK timestamps are month-level only, not daily; monthly aggregation was applied uniformly.
- Boston extraction-window terminus produced a partial final bin containing only the first 10 days of April 2026 (resolved by §3.4 partial-bin exclusion).

Most of these issues would have produced silent failures (silently incorrect numbers) if undetected; the two PostgreSQL configuration issues would have produced loud failures (connection or authentication errors). Each was caught by a pre-patch audit or validation step before any visibly wrong numbers entered downstream analyses.

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### Declaration of generative AI use in the writing process

During the preparation of this work the author used Anthropic’s Claude (claude.ai) for editorial review, structural reorganization, citation format conversion, adversarial audit of methodological claims against the underlying data, and LaTeX compile-error correction. The author reviewed and edited all content produced with the aid of this tool and takes full responsibility for the content of the publication.

### References

- Andresen, M. A. (2009). Testing for similarity in area-based spatial patterns: A nonparametric Monte Carlo approach. *Applied Geography*, 29(3), 333–345.
- Bowers, K. (2014). Risky facilities: Crime radiators or crime absorbers? A comparison of internal and external levels of theft. *Journal of Quantitative Criminology*, 30(3), 389–414.
- Braga, A. A., Turchan, B. S., Papachristos, A. V., & Hureau, D. M. (2019). Hot spots policing and crime reduction: An update of an ongoing systematic review and meta-analysis. *Journal of Experimental Criminology*, 15(3), 289–311.
- Brantingham, P. J., & Brantingham, P. L. (1984). *Patterns in crime*. New York: Macmillan.

- Bühlmann, H., & Gisler, A. (2005). *A course in credibility theory and its applications*. Universitext. Berlin: Springer.
- Cameron, A. C., & Miller, D. L. (2015). A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2), 317–372.
- Cameron, A. C., & Trivedi, P. K. (2013). *Regression analysis of count data* (2nd ed.). Cambridge: Cambridge University Press.
- Chainey, S., Tompson, L., & Uhlig, S. (2008). The utility of hotspot mapping for predicting spatial patterns of crime. *Security Journal*, 21(1–2), 4–28.
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). New York: ACM.
- Czado, C., Gneiting, T., & Held, L. (2009). Predictive model assessment for count data. *Biometrics*, 65(4), 1254–1261.
- Davison, A. C., & Hinkley, D. V. (1997). *Bootstrap methods and their application*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge: Cambridge University Press.
- Eck, J. E., Clarke, R. V., & Guerette, R. T. (2007). Risky facilities: Crime concentration in homogeneous sets of establishments and facilities. *Crime Prevention Studies*, 21, 225–264.
- Federal Bureau of Investigation. (2021). *National Incident-Based Reporting System (NIBRS) user manual*. Washington, DC: U.S. Department of Justice.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378.
- Groff, E. R., Weisburd, D., & Yang, S.-M. (2010). Is it important to examine crime trends at a local “micro” level?: A longitudinal analysis of street to street variability in crime trajectories. *Journal of Quantitative Criminology*, 26(1), 7–32.
- Hardyns, W., & Khalfa, R. (2023). Predicting crime across cities and regions: A comparative analysis of predictive modelling in three Belgian settings. *Applied Spatial Analysis and Policy*, 16(1), 485–508.
- Heaven, W. D. (2020, July 17). Predictive policing algorithms are racist. They need to be dismantled. *MIT Technology Review*. Retrieved April 2026 from <https://www.technologyreview.com/2020/07/17/1005396/predictive-policing-algorithms-racist-dismantled-machine-learning-bias-criminal-justice/>
- Hilbe, J. M. (2011). *Negative binomial regression* (2nd ed.). Cambridge: Cambridge University Press.

- Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, 2(1), 193–218.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30* (pp. 3146–3154).
- Khalifa, R., & Hardyns, W. (2025). Comparing machine learning-based crime predictions across micro-geographic units: Street segments, rectangular grids, and hexagonal grids. *Applied Spatial Analysis and Policy*, 18, 79.
- Khalifa, R., Snaphaan, T., Ristea, A., Kounadi, O., & Hardyns, W. (2025). Predicting crime at micro places: Comparing machine learning methods across European cities. In M. Leitner (Ed.), *New research in crime modeling and mapping using geospatial technologies* (pp. 113–143). Geotechnologies and the Environment, vol. 27. Cham: Springer.
- Khalifa, R., Theinert, N., & Hardyns, W. (2025). Comparing XAI techniques for interpreting short-term burglary predictions at micro-places. *Computational Urban Science*, 5(1), 27.
- Kontur. (2024). Kontur population dataset. Retrieved April 2026 from <https://data.humdata.org/dataset/kontur-population-dataset>
- Kounadi, O., Ristea, A., Araujo, A., Jr., & Leitner, M. (2020). A systematic review on spatial crime forecasting. *Crime Science*, 9(1), 7.
- Kronkvist, K., Borg, A., Boldt, M., & Gerell, M. (2025). Predicting public violent crime using register and OpenStreetMap data: A risk terrain modeling approach across three cities of varying size. *Applied Spatial Analysis and Policy*, 18(1), 9.
- Lum, K., & Isaac, W. (2016). To predict and serve? *Significance*, 13(5), 14–19.
- McCarthy, A., Chen, H., Fox, B., & Verona, E. (2025). The empirical clustering of crimes. *Journal of Criminal Justice*, 101, 102526.
- Mohler, G. (2014). Marked point process hotspot maps for homicide and gun crime prediction in Chicago. *International Journal of Forecasting*, 30(3), 491–497.
- Mohler, G. O., Short, M. B., Brantingham, P. J., Schoenberg, F. P., & Tita, G. E. (2011). Self-exciting point process modeling of crime. *Journal of the American Statistical Association*, 106(493), 100–108.
- Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning* (pp. 625–632). New York: ACM.
- Richardson, R., Schultz, J. M., & Crawford, K. (2019). Dirty data, bad predictions: How civil rights violations impact police data, predictive policing systems, and justice. *New York University Law Review Online*, 94, 192–233.



- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20, 53–65.
- Rummens, A., & Hardyns, W. (2021). The effect of spatiotemporal resolution on predictive policing model performance. *International Journal of Forecasting*, 37(1), 125–133.
- Sherman, L. W., Gartin, P. R., & Buerger, M. E. (1989). Hot spots of predatory crime: Routine activities and the criminology of place. *Criminology*, 27(1), 27–56.
- Steenbeek, W., & Weisburd, D. (2016). Where the action is in crime? An examination of variability of crime across different spatial units in The Hague, 2001–2009. *Journal of Quantitative Criminology*, 32(3), 449–469.
- Tibshirani, R., Walther, G., & Hastie, T. (2001). Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63(2), 411–423.
- Uber Technologies. (n.d.). H3: Hexagonal hierarchical spatial index. Retrieved April 2026 from <https://h3geo.org>
- UK Home Office. (n.d.). *data.police.uk*. Retrieved April 2026 from <https://data.police.uk>
- Weisburd, D. (2015). The law of crime concentration and the criminology of place. *Criminology*, 53(2), 133–157.
- Wheeler, A. P., & Steenbeek, W. (2021). Mapping the risk terrain for crime using machine learning. *Journal of Quantitative Criminology*, 37(2), 445–480.